

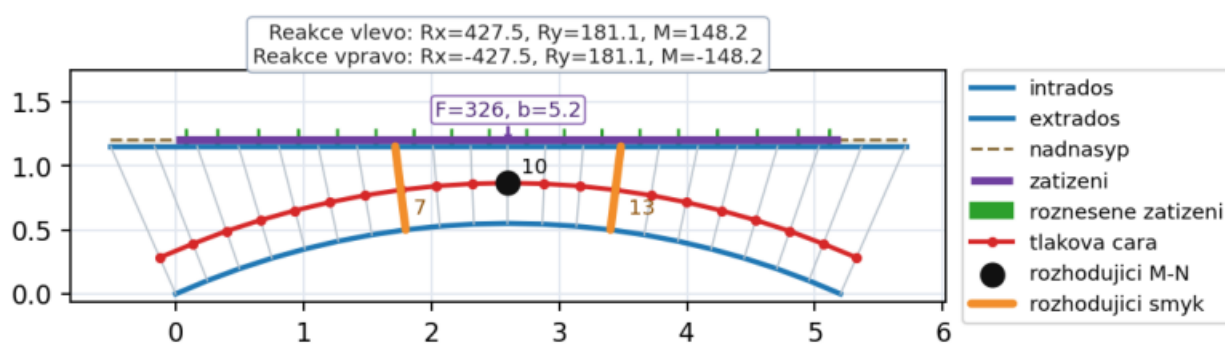
ArchLBT vypocetni protokol

1) Zadani ulohy

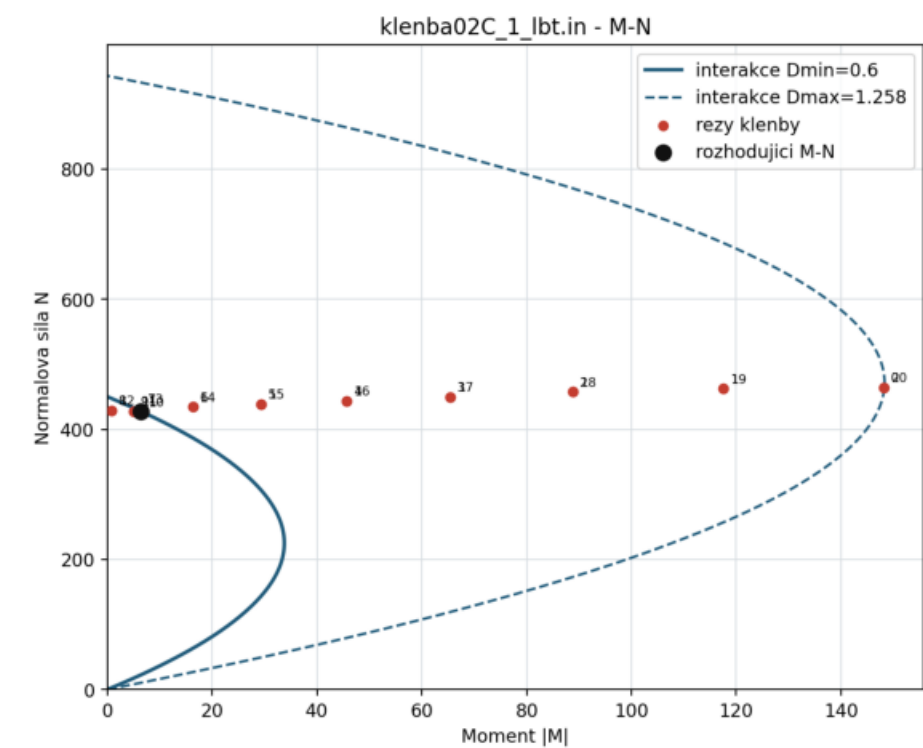
```
# ArchLBT input
# Unit convention: use one consistent force-length system.
# Geometry is in length units, typical m. Unit weights are force/volume, typical kN/m3.
# Surface point_load force F is the total force on the loaded length, typical kN.
# Self-weight and fill loads are multiplied internally by arch_width; point_load F is not.
span = 5.2
rise = 0.55
geom_code = 1
# extrados_mode: normal, horizontal, horizontal_width_radial_joints.
extrados_mode = horizontal_width_radial_joints
thickness = 0.6
# arch_width is the out-of-plane width b of the analysed arch strip.
arch_width = 0.5
masonry_unit_weight = 24.3
block_count = 20
point_load = 2.6 326 5.2
fill_height = 0.05
fill_unit_weight = 21.76
q_code = 1
fill_spread_angle_deg = 0
friction_coefficient = 0.6
compression_strength = 1500
```

2) Obrazek Klenba

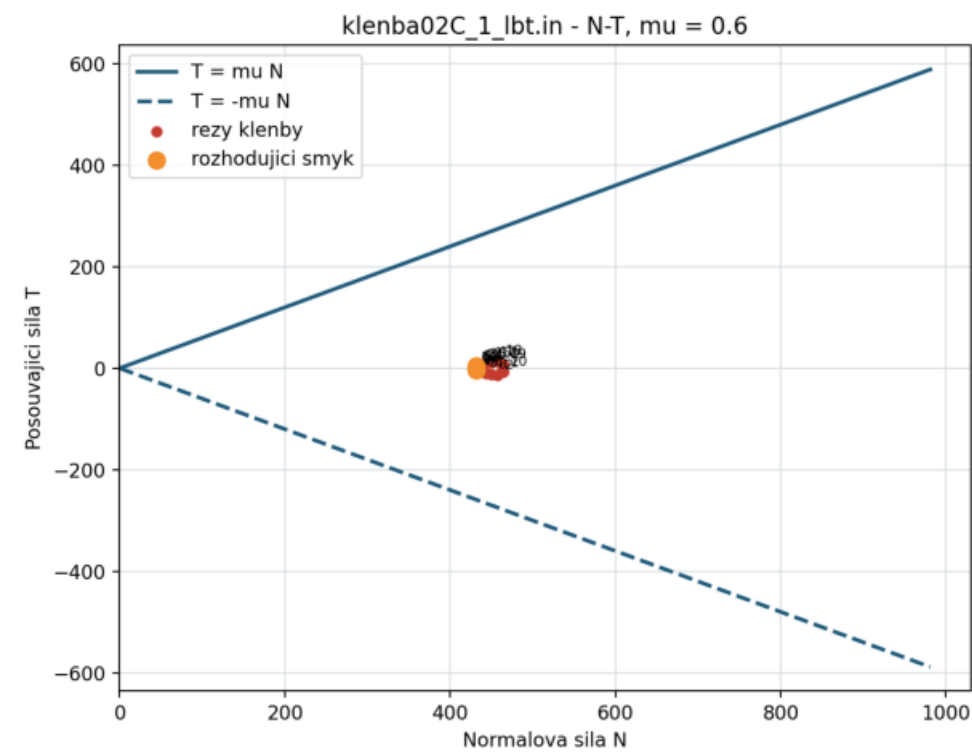
klenba02C_1_lbt.in - $\lambda = 0.927509$



3) Graf M-N



4) Graf N-T



5) Vysledky

rez	xi	yi	xe	ye	N	M	T	e = M/N
0	-0	0	-0.50933024	1.15	464.24456	-148.20768	7.537699	-0.31924483
1	0.24694569	0.10326891	-0.1653625	1.15	462.53572	-117.50781	-7.7073643	-0.25405132
2	0.49798112	0.19615515	0.16748353	1.15	457.33457	-88.841671	-14.055872	-0.19425969
3	0.75266999	0.27849727	0.49084436	1.15	449.92585	-65.386407	-11.718414	-0.14532707
4	1.0105696	0.35015215	0.80620009	1.15	443.71887	-45.636705	-9.6109476	-0.10285049
5	1.2712318	0.41099525	1.1149038	1.15	438.61833	-29.331899	-7.702183	-0.0668734
6	1.5342034	0.46092083	1.4182068	1.15	434.54648	-16.26163	-5.9601122	-0.037422073
7	1.7990275	0.49984212	1.7172795	1.15	431.44153	-6.2603804	-4.3521239	-0.014510379
8	2.0652436	0.52769145	2.0132312	1.15	429.25655	0.79665528	-2.8451227	0.0018558955
9	2.3323893	0.54442044	2.3071262	1.15	427.95858	4.9965882	-1.405649	0.011675401
10	2.6	0.55	2.6	1.15	427.52808	6.3908903	-2.920622e-09	0.014948469
11	2.8676107	0.54442044	2.8928738	1.15	427.95858	4.9965882	1.405649	0.011675401
12	3.1347564	0.52769145	3.1867688	1.15	429.25655	0.79665531	2.8451227	0.0018558955
13	3.4009725	0.49984212	3.4827205	1.15	431.44153	-6.2603803	4.3521239	-0.014510379
14	3.6657966	0.46092083	3.7817932	1.15	434.54648	-16.26163	5.9601122	-0.037422073
15	3.9287682	0.41099525	4.0850962	1.15	438.61833	-29.331899	7.702183	-0.0668734
16	4.1894304	0.35015215	4.3937999	1.15	443.71887	-45.636705	9.6109475	-0.10285049
17	4.44733	0.27849727	4.7091556	1.15	449.92585	-65.386406	11.718414	-0.14532707
18	4.7020189	0.19615515	5.0325165	1.15	457.33457	-88.841671	14.055872	-0.19425969
19	4.9530543	0.10326892	5.3653625	1.15	462.53572	-117.50781	7.7073643	-0.25405132
20	5.2	7.0038708e-10	5.7093302	1.15	464.24456	-148.20768	-7.537699	-0.31924483

6) Report

ArchLBT prepared model
input = D:\Documents\2026\MASARCH2026\python\klenba02C_1_lbt.in

Geometry
span = 5.2
rise = 0.55
thickness D = 0.6
section depth min/max = 0.6 / 1.25774
arch width b = 0.5
geom_CODE = 1
extrados mode = horizontal_width_radial_joints
blocks = 20
joints = 21

Loads
arch self-weight vertical = 56.462087
backfill vertical = 3.382951
variable vertical = 326.000000
total vertical = 385.845039

Lower-bound skeleton
state = permanent + lambda * variable
solution = statically admissible reactions Rx, Ry, M0
constraints = 0 <= N <= fc*b*D_i and abs(M) <= N*(D_i/2 - N/(2*fc*b))
shear constraint = abs(T) <= mu*N
fc = 1500
b = 0.5
Nmax min/max = 450 / 943.307
mu = 0.6
status = optimal
lambda_max = 0.927509
governing joint = 10
governing M-N joints = 10
governing shear joints = 7, 13
governing compression joints = 10
governing tension joints = 10
Rx = 427.528082
Ry = 181.106514
M0 = 148.207676

Prepared arrays
intrados/extrados = 21 x 4
local coordinates = 21 x 4
masonry loads = 21 x 3
backfill loads = 21 x 3
variable loads = 21 x 3

Global equilibrium check
moment origin = [0, 0]
loads sum Fx = 0.000000000e+00

6) Report / pokracovani

loads sum Fy = -3.622130288e+02
loads sum M = -9.417538748e+02
left reaction = [4.275280824e+02, 1.811065144e+02], M = 1.482076765e+02
right reaction = [-4.275280824e+02, 1.811065143e+02], M = -1.482076763e+02
residual sum Fx = 0.000000000e+00
residual sum Fy = 0.000000000e+00
residual sum M = 0.000000000e+00

7) Generovana pole

local coordinates = 21 x 4
[[0.62887147 1.98772676 0.29993464 0.17301424]
[0.56250422 1.94603372 0.29561183 0.15776206]
[0.5047397 1.90434068 0.29194249 0.14149623]
[0.45499167 1.86264763 0.28885815 0.12435149]
[0.41277218 1.82095459 0.28630378 0.10645717]
[0.37767924 1.77926155 0.28423557 0.08793759]
[0.34938708 1.7375685 0.2826192 0.0689127]
[0.32763852 1.69587546 0.28142861 0.04949872]
[0.31223918 1.65418241 0.28064499 0.02980895]
[0.30305314 1.61248937 0.28025618 0.00995456]
[0.3 1.57079633 0.28025618 -0.00995456]
[0.30305314 1.52910328 0.28064499 -0.02980895]
[0.31223918 1.48741024 0.28142861 -0.04949872]
[0.32763852 1.4457172 0.2826192 -0.0689127]
[0.34938708 1.40402415 0.28423557 -0.08793759]
[0.37767924 1.36233111 0.28630378 -0.10645717]
[0.41277218 1.32063806 0.28885815 -0.12435149]
[0.45499167 1.27894502 0.29194249 -0.14149623]
[0.5047397 1.23725198 0.29561183 -0.15776206]
[0.56250422 1.19555893 0.29993464 -0.17301424]
[0.62887147 1.15386589 1. -0.41693044]]

masonry loads = 21 x 3
[[0. 8.46402104 1.17408834]
[0. 7.51535603 1.04314094]
[0. 6.70628215 0.93059474]
[0. 6.0234587 0.83534935]
[0. 5.45605572 0.75630299]
[0. 4.99537253 0.69243995]
[0. 4.63454427 0.64288399]
[0. 4.3683188 0.60693078]
[0. 4.19289125 0.58406836]
[0. 4.10578702 0.57399252]
[0. 4.10578702 0.57662264]
[0. 4.19289125 0.59212282]
[0. 4.3683188 0.62093336]
[0. 4.63454426 0.66381833]
[0. 4.99537252 0.72193623]
[0. 5.45605571 0.79694308]
[0. 6.02345869 0.89114064]
[0. 6.70628214 1.00768742]
[0. 7.51535602 1.1508976]
[0. 8.46402103 1.32666349]
[0. 0. 0.]]

backfill loads = 21 x 3
[[0. 0.3742369 -0.03094237]
[0. 0.36213648 -0.01438807]
[0. 0.35181658 -0.00125542]
[0. 0.34310703 0.00918328]
[0. 0.33586969 0.01752135]
[0. 0.32999358 0.0242504]
[0. 0.32539114 0.02978567]
[0. 0.32199538 0.0344863]
[0. 0.31975777 0.03867191]
[0. 0.31864674 0.04263665]
[0. 0.31864674 0.04666165]
[0. 0.31975777 0.05102664]
[0. 0.32199538 0.05602142]
[0. 0.32539114 0.06195784]
[0. 0.32999358 0.06918308]
[0. 0.33586969 0.07809502]
[0. 0.34310703 0.08916069]
[0. 0.35181658 0.10293915]
[0. 0.36213648 0.12011046]
[0. 0.3742369 0.14151318]
[0. 0. 0.]]

variable loads = 21 x 3
[[0. 0. -0.]
[0. 10.49992922 0.45097376]

7) Generovana pole / pokracovani

[0.	20.27223656	-0.07233905]
[0.	19.77037818	0.52915516]
[0.	19.353351	1.00960846]
[0.	19.01476024	1.39734716]
[0.	18.74956038	1.71629827]
[0.	18.55389123	1.98715608]
[0.	18.42495632	2.22833749]
[0.	18.36093686	2.45679199]
[0.	18.36093686	2.68871904]
[0.	18.42495632	2.94023706]
[0.	18.55389123	3.22804423]
[0.	18.74956038	3.57011017]
[0.	19.01476024	3.98644028]
[0.	19.353351	4.49996072]
[0.	19.77037818	5.13758244]
[0.	20.27223655	5.93151911]
[0.	10.49992925	2.61438328]
[0.	0.	0.]
[0.	0.	0.]]